

Rutv Kocheta

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EDUCATION

Indian Institute of Technology, Hyderabad <i>B.Tech in Mathematics and Computing, Minor in Artificial Intelligence</i>	2021 - 2025 CGPA: 9.81
Class X & XII <i>Class X (CBSE): 97.6% Class XII (Telangana State Board): 97.7%</i>	2018 - 2021

EXPERIENCE

Software Engineer <i>D.E. Shaw and Co.</i>	June 2025 - Present Hyderabad
- Converted a Summer 2024 internship into a full-time Software Engineer role on the Private Credit Team. - Own the Private Credit portfolio dashboard (Market Value, PnL, Buffer, Capital Allocation) used daily by the entire desk, built on the firm's in-house caching infrastructure for live data refresh; extended it with a natural-language chatbot using Claude Code Agent SDK for ad-hoc querying. - Built a data visualization tool from scratch to automate plotting of domain-specific charts and tables, cutting analyst turnaround time by 98% during the internship.	
Founder's Office Intern <i>Locus.sh</i>	May 2023 - July 2023 Bengaluru
Worked under the CEO and Chief of Staff to optimize last-mile delivery solutions; conducted analysis on geo-coding and route optimization software, designing CNN-based models and mathematical algorithms to improve route planning for clients.	
Teaching Assistant <i>Probability and Random Variables, Math Foundation, Introduction to Programming</i>	2024, 2023, 2022
- Worked under Prof. Panolan and Prof. Kataoka for Introduction to Programming, Prof. Neeraj Kumar for Math Foundation, and Prof. PN Karthik for Probability and Random Variables. - Graded and set papers and assignments. Took bi-weekly tutorial sessions.	

TECHNICAL SKILLS

Languages: Python, C++, C, Bash, JavaScript
Development: Node.js, Flask, FastAPI, PostgreSQL, MongoDB, SQLite, MySQL, React, Redux Toolkit, Git
Data Science: PyTorch, TensorFlow, OpenCV, Pandas, NumPy, SciPy, Matplotlib

SELECTED PROJECTS

Quantitative Crypto Trading <i>Inter-IIT Tech Meet 12.0, IIT Madras</i>	December 2023
Built a back-testing engine from scratch using object-oriented principles to evaluate strategies; developed algorithmic and ML-based trading strategies for the crypto currency market, securing a top-10 position.	
Multi-Agent Multi-Armed Bandits <i>Guide: Prof. Ganesh Ghalme</i>	Jan - May 2024
Minimizing Access and Repair Costs for Reed-Solomon Codes <i>Guide: Prof. Myna Vajha</i>	Jan - May 2025
Nearest Neighbours in Metric Spaces via Kd-trees and R-trees <i>Course Project</i>	November 2022
Implemented Kd-tree structures and heuristics for nearest-neighbour search in circular ranges using pruning techniques and median-finding algorithms.	
Shidoku Solver using Gröbner Basis <i>Course Project</i>	May 2024
Modeled a 4×4 Shidoku board as a system of polynomial equations; enumerated all valid completions of a partially filled board using Gröbner basis methods.	

ACHIEVEMENTS

- Secured All India Rank 970 in JEE Advanced amongst 2,45,000 candidates (2021).
- Received the KVPY SX fellowship, All India Rank 966, from the Government of India (2021).
- Qualified for the Indian National Mathematics Olympiad amongst 300 students nationwide (2018).

RELEVANT COURSEWORK

Mathematics: Combinatorics and Graph Theory, Number Theory, Probability Theory, Linear Algebra, Real Analysis, Numerical Analysis, Applied Statistics, Ordinary Differential Equations, Functional Analysis, Modules and Fields

Computer Science: Data Structures and Applications, Algorithms, Database Management Systems, Theory of Computation, Operating Systems

Minor (Artificial Intelligence): Foundations of Machine Learning, Deep Learning, Game Theory, Reinforcement Learning